

IRSIM: A Switch-Level Circuit Simulator

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Regular readers of this section must have seen many circuit simulators by now, but how many of these actually allow us to modify the circuit at the simulation stage? Well, here we have one such simulator, IRSIM. This tool allows us to modify the circuit under simulation and re-simulate it using IRSIM. IRSIM is available for both Linux and Windows operating systems (OSes) in the DVD accompanying this month's issue of EFY Plus.

IRSIM, not the Invisionix Systems' portable instant messenger you get when you Google the term, is an interactive and incremental logic-level simulator used for metal oxide semiconductor (MOS) transistor based digital circuits. Pronounced as *eye-are-sim*, this tool is not an infra-red simulator, but is always referred to as a switch-level simulator.

Why switch-level simulator

The open source software IRSIM has the ability to simulate in two modes: switch and linear. During switch mode, it initialises and determines

Quick view

- Version: 9.7.87
- Licence: GNU public licence (GPL)
- Key drawbacks:
 1. Cannot remove or reload a file being simulated
- Key features:
 1. Event-driven circuit simulator
 2. Faster than Spice
 3. Has a built-in display for interactive interactions

Competitors

- ModelSim
- Spectre
- SPICE

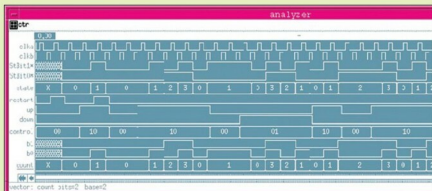


Fig. 1: IRSIM simulation of the 2-bit up-down counter

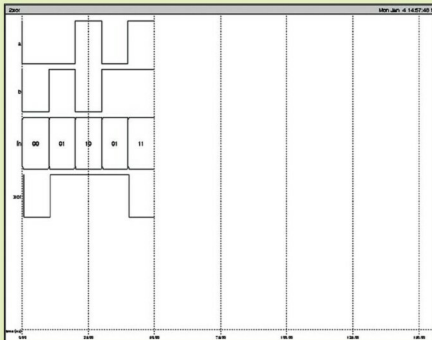


Fig. 2: IRSIM simulation output file created using Print Menu command

the functionality of a MOS network, whereas in case of linear mode, it determines the number of times the gate delays and the circuit transition occurs.

An interesting point to note is that although it has the ability to work in two different modes, we still relate it to only one mode. The reason is that, IRSIM models the circuits at transistor level where these transistors are

modelled as switches. At this point, most analogue and higher order properties of the devices into consideration are ignored and these are treated as ideal on/off connections.

A simulator but not an editor

We are all aware of the importance of a digital circuit simulator and that it needs a netlist to work. A netlist is a circuit schematic with components